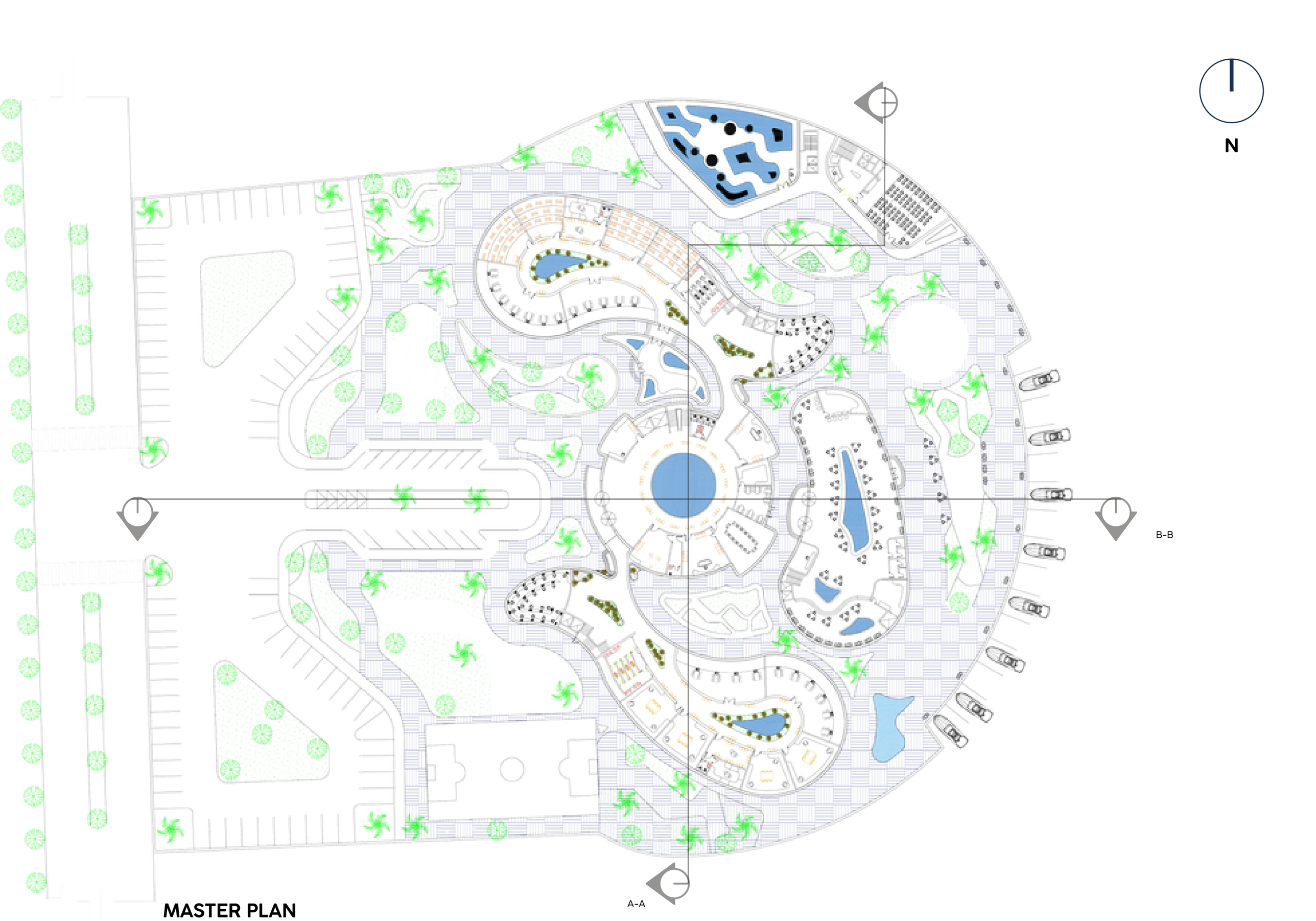
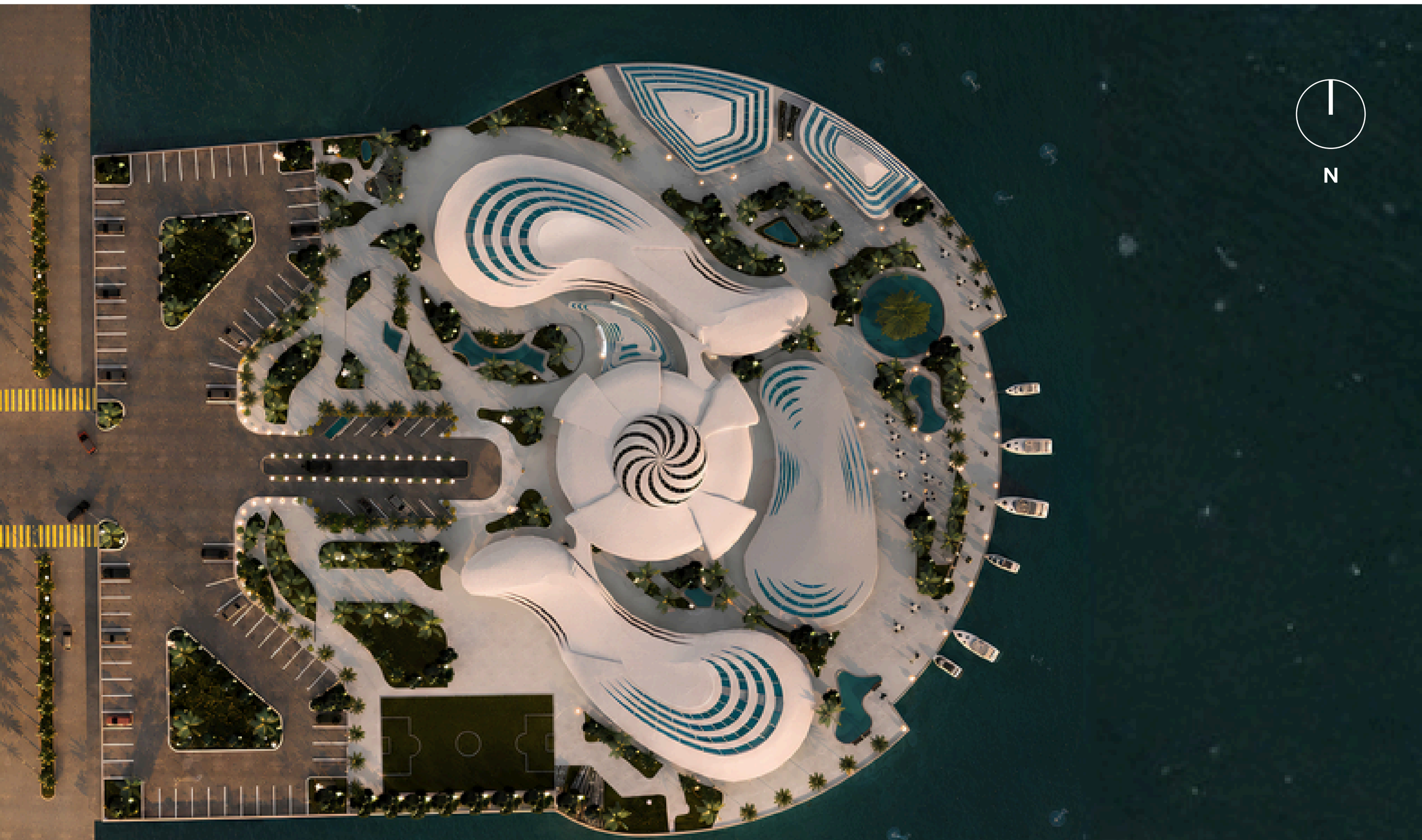
وُسُحُبٌ

﴿أَوْ كَظُلُمَاتٍ فِي بَحْرٍ لُجِّيٍّ يَغْشَاهُ مَوْجٌ مِنْ فَوْقِهِ مَوْجٌ مِنْ فَوْقِهِ سَحَابٌ﴾

سورة النور — الآية 40

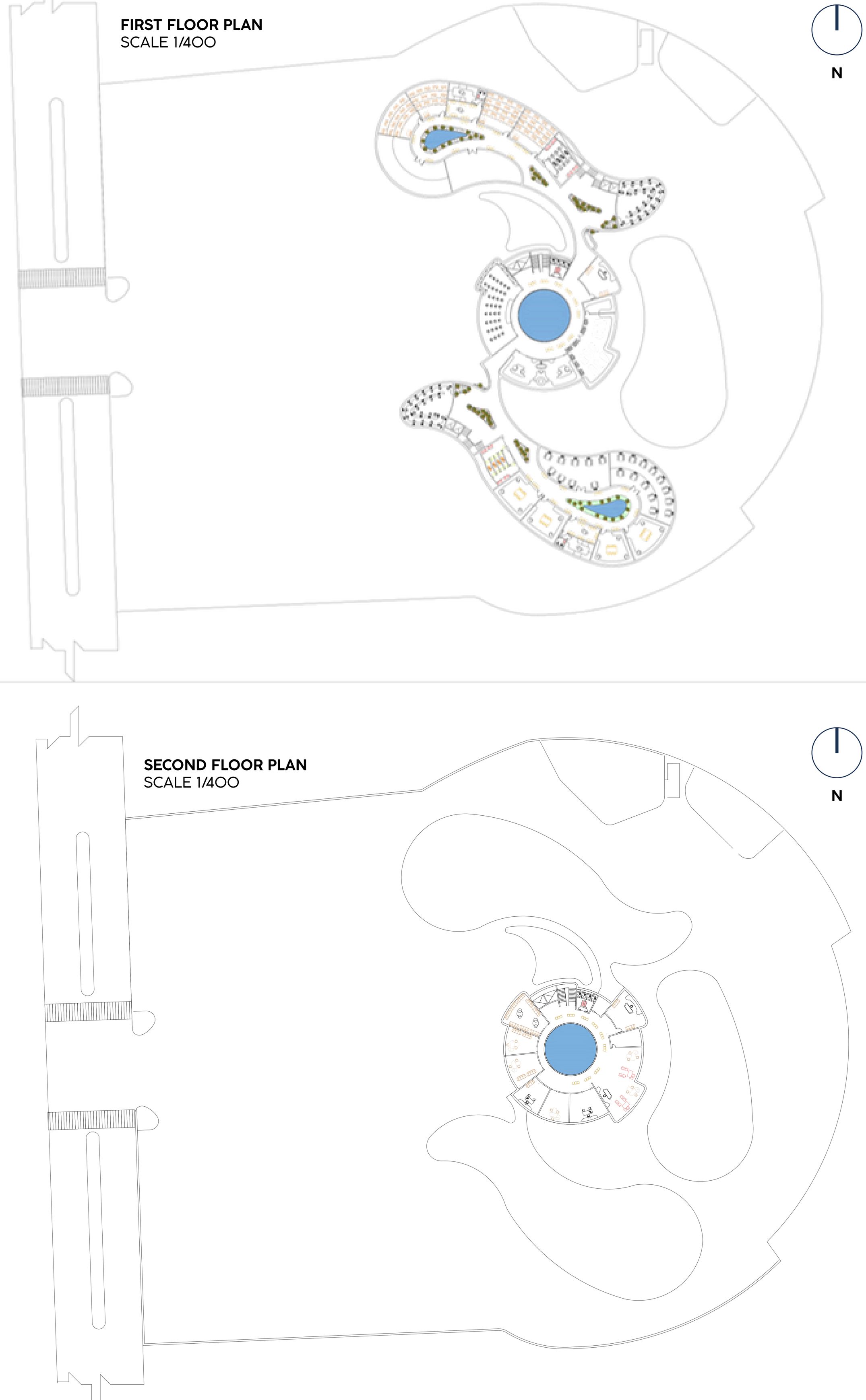


MASTER PLAN
SCALE 1/200



LAYOUT
SCALE 1/200

FIRST FLOOR PLAN
SCALE 1/400



SECOND FLOOR PLAN
SCALE 1/400

لُجّة

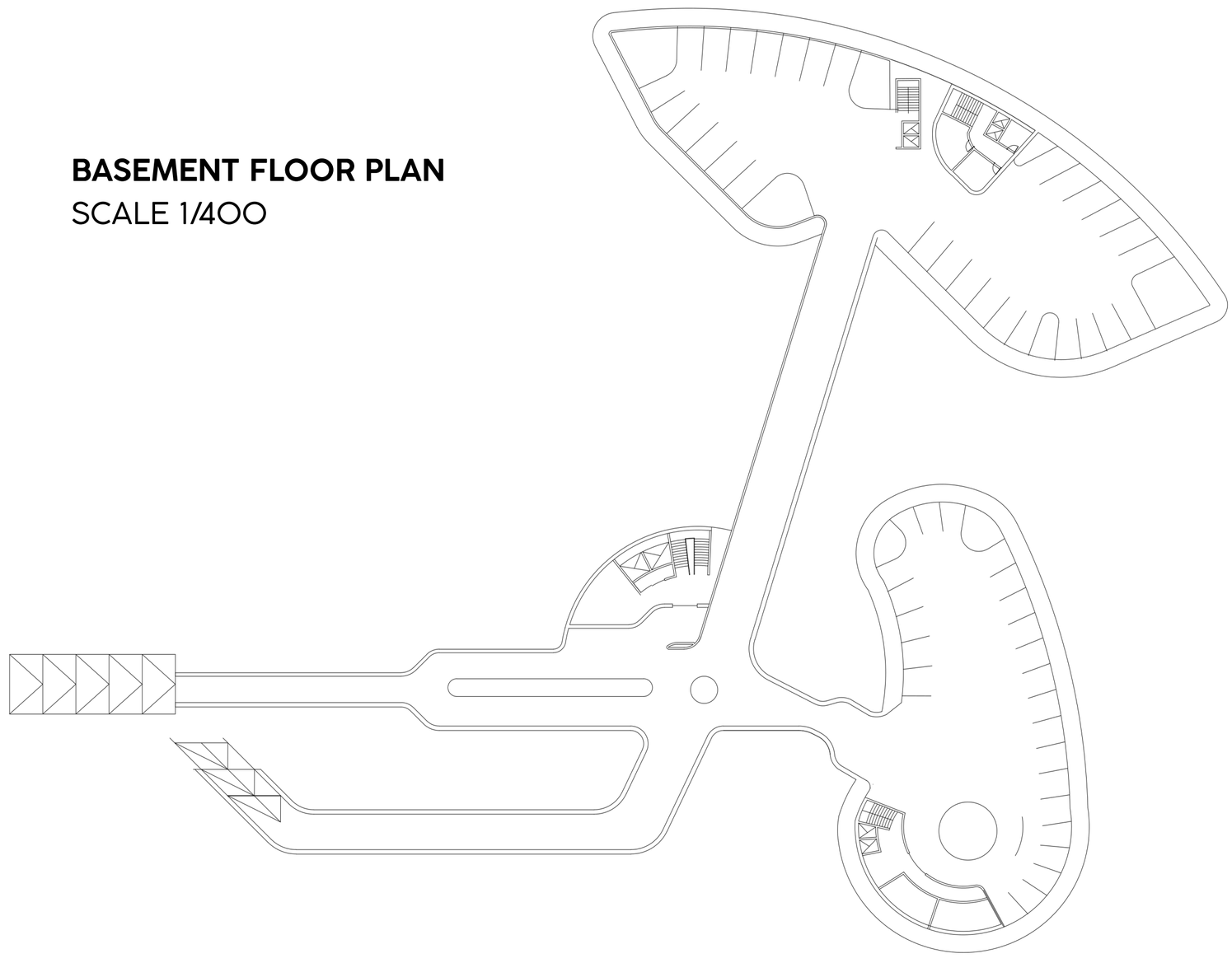
SOUTH ELEVATION

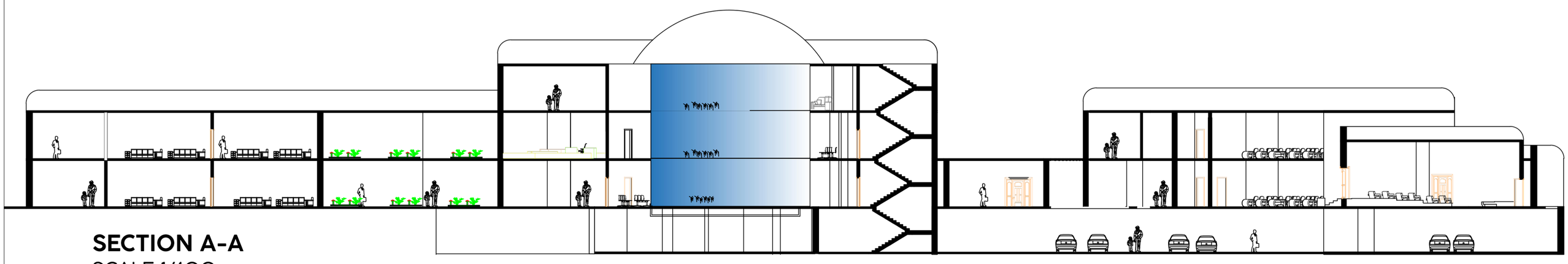
WEST ELEVATION

NORTH ELEVATION

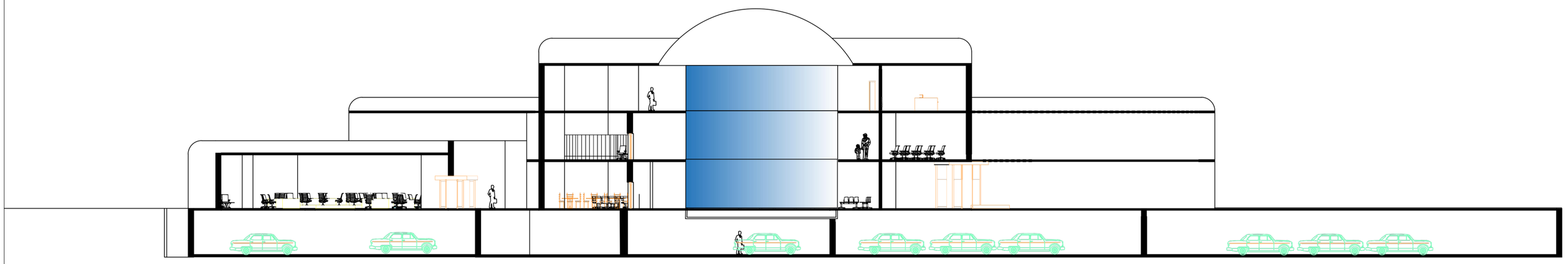
NORTH ELEVATION

BASEMENT FLOOR PLAN
SCALE 1/400





SECTION A-A
SCALE 1/400



SECTION B-B
SCALE 1/400

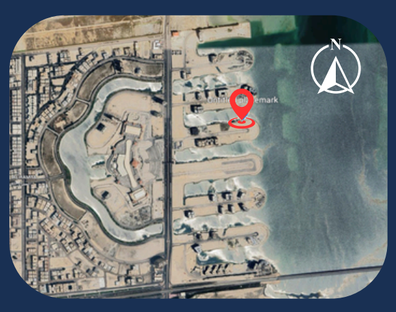
SITE ANALYSIS FOR SCHOOL OF MARINE SCIENCE & INNOVATION

BY: REEM AL-HARKAN
SUPERVISED BY: DR.SHAIMAA ASHOUR

LOCATION:

SAUDI ARABIA

KHOBAR

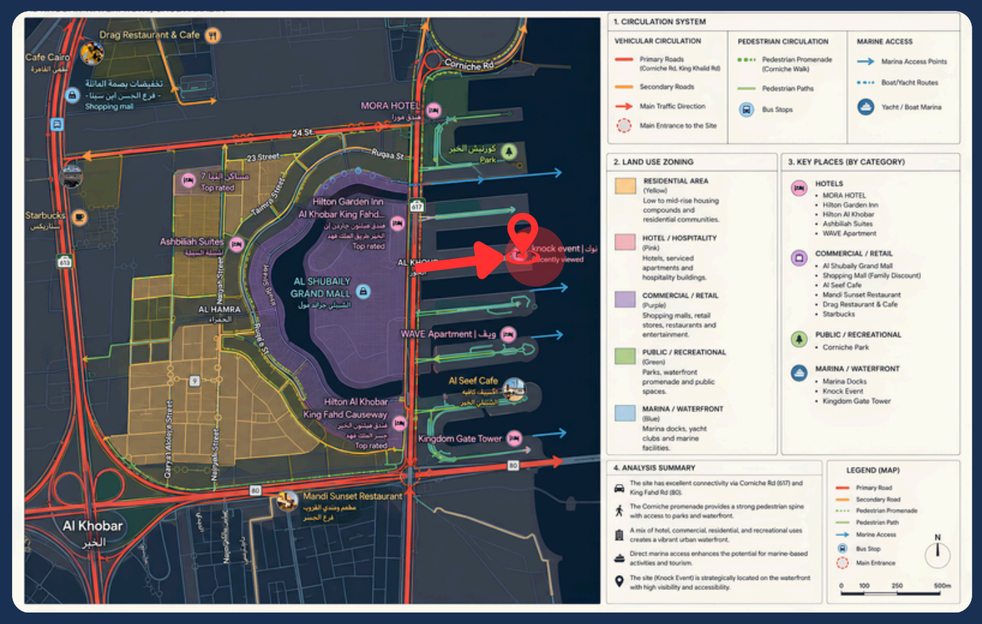


AL KHOUR

SITE BOUNDARIES

AL KHOBAR IS A VIBRANT COASTAL CITY KNOWN FOR ITS BEAUTIFUL WATERFRONT, MODERN LIFESTYLE, AND TOURIST ATTRACTIONS, MAKING IT ONE OF SAUDI ARABIA'S MOST POPULAR DESTINATIONS.

CIRCULATION



NEIGHBORING AREAS:

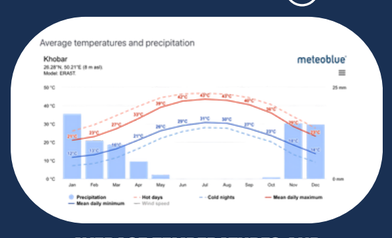
AL SHUBAILY GRAND MALL
4 KM DRIVE AWAY

HILTON HOTIL
4 KM DRIVE AWAY

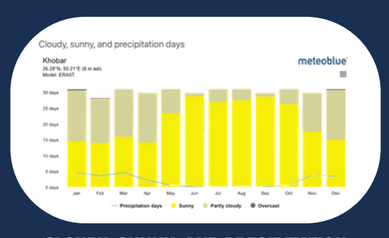
AYDAH AL-HARTHI MOSQUE
5.4 KM DRIVE AWAY

DR.SULAIMAN AL HABIB MEDICAL CENTER
9.5 KM DRIVE AWAY

CLIMATE :

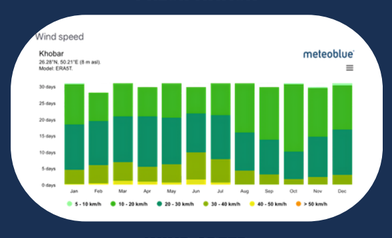


AVERAGE TEMPERATURES AND PRECIPITATION

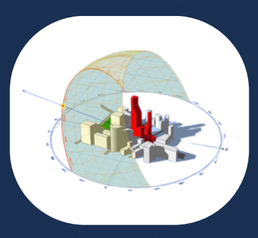


CLOUDY, SUNNY, AND PRECIPITATION DAYS

WIND ROSE



WIND SPEED



SUN PATH IN JANUARY

SUN PATH IN AUGUST

STRENGTHS

- PRIME WATERFRONT LOCATION.
- EXCELLENT ROAD ACCESSIBILITY.
- SURROUNDED BY HOTELS AND RETAIL.
- ACTIVE PEDESTRIAN PROMENADE.

WEAKNESSES

- TRAFFIC CONGESTION.
- LIMITED PARKING.
- HIGH HUMIDITY AND SALT EXPOSURE.
- STRONG SOLAR HEAT.

OPPORTUNITIES

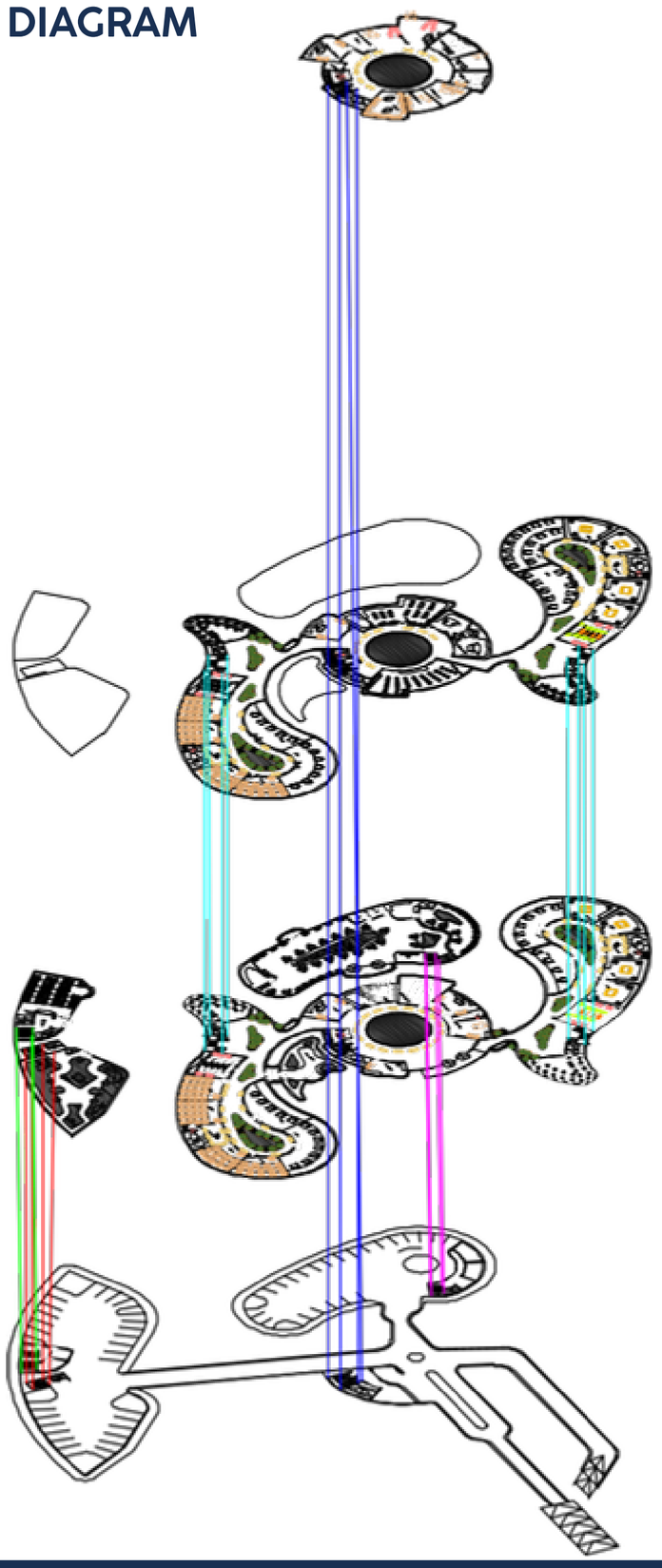
- ENHANCE MARINE TOURISM.
- CREATE A LANDMARK DESTINATION.
- STRENGTHEN PUBLIC WATERFRONT SPACES.
- APPLY SUSTAINABLE DESIGN.

THREATS

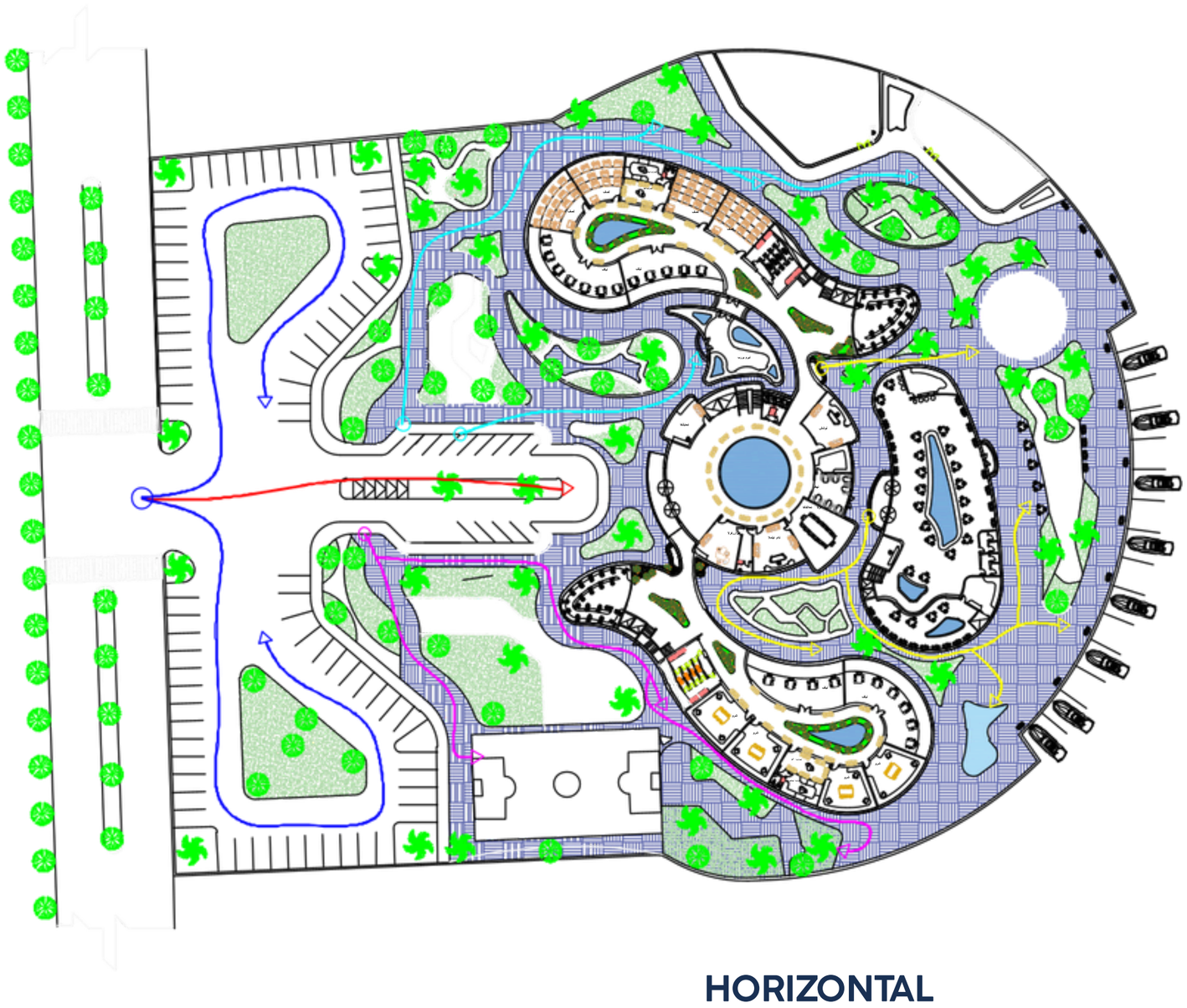
- COASTAL FLOODING RISKS.
- NEARBY COMMERCIAL COMPETITION.
- SEASONAL OVERCROWDING.
- SALT CORROSION AND MAINTENANCE.

KHOBAR IS CHARACTERIZED BY A HOT DESERT CLIMATE WITH EXTREMELY HOT SUMMERS, MILD WINTERS, HIGH HUMIDITY DUE TO ITS COASTAL LOCATION, LIMITED ANNUAL RAINFALL, ABUNDANT SUNSHINE, AND PREDOMINANTLY NORTHWESTERLY WINDS. THESE CLIMATIC CONDITIONS HIGHLIGHT THE IMPORTANCE OF PASSIVE DESIGN STRATEGIES, INCLUDING SOLAR SHADING, NATURAL VENTILATION, AND HUMIDITY CONTROL, TO IMPROVE THERMAL COMFORT AND ENERGY EFFICIENCY.

VERTICAL CIRCULATION DIAGRAM



N

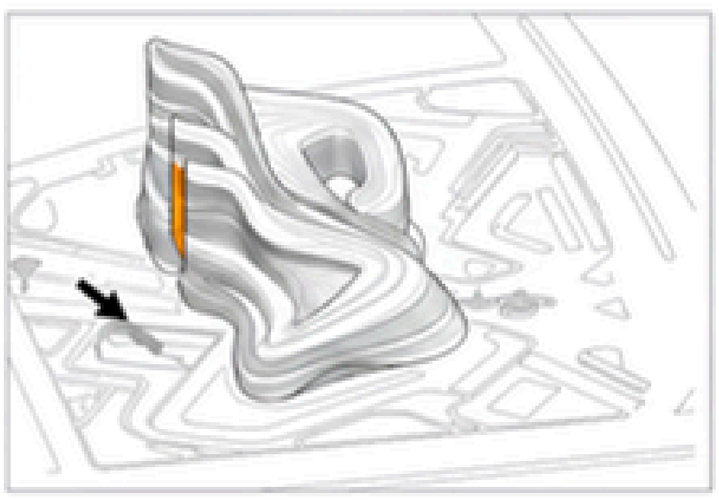


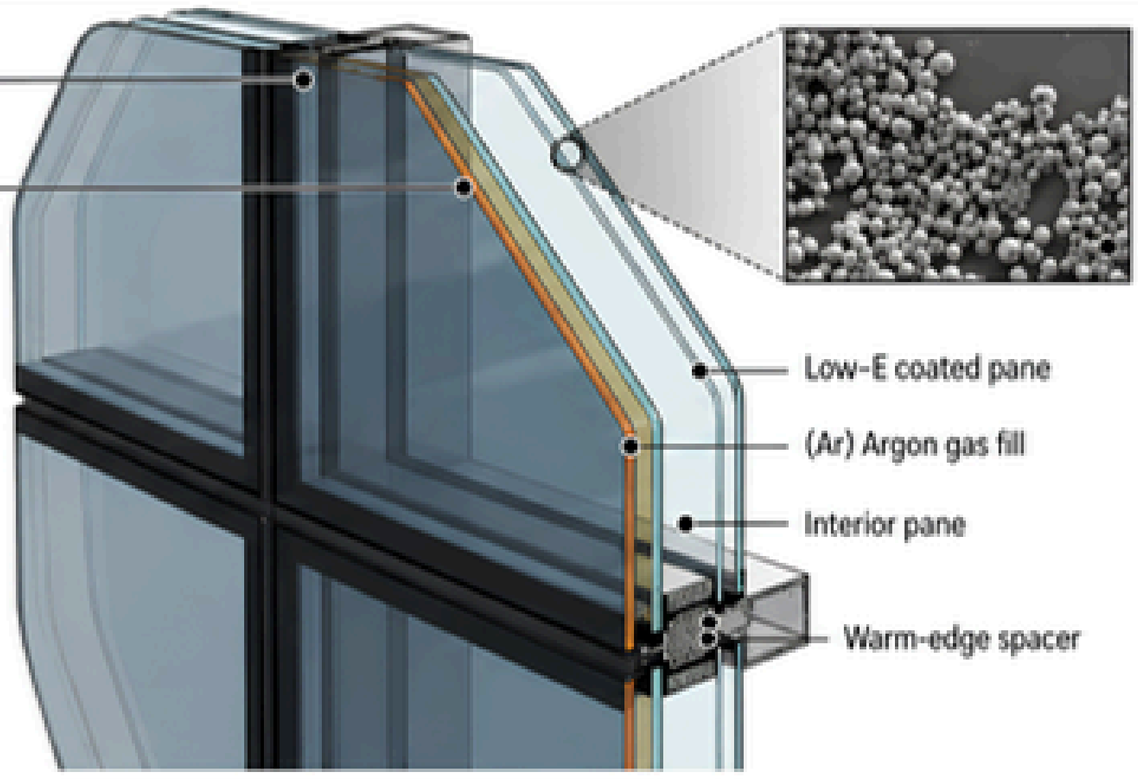
HORIZONTAL CIRCULATION DIAGRAM

FUTURISTIC DOUBLE-SKIN FAÇADE SYSTEM – THERMAL GAIN REDUCTION – FUTURE COMPLEX

Curved reflective outer façade with high reflection and low absorption

Low-E coated double glazing unit with argon gas fill

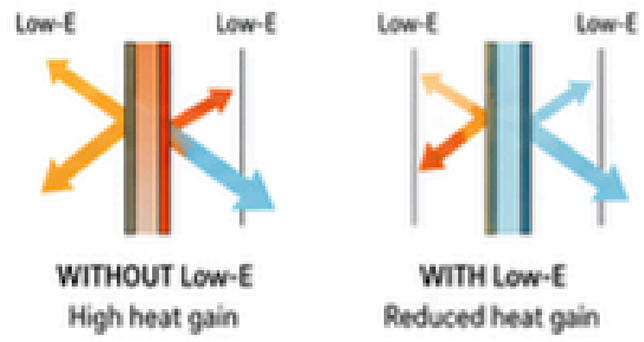




DESICCANT COATED WITH ENERGY REFLECTING MATERIAL
Reflects heat outward and allows natural light to enter

ALLOWS VISIBLE LIGHT TRANSMISSION (IR)

Infrared (IR) Visible Light



WATER SUSTAINABILITY STRATEGIES

LUJJA – SCHOOL OF MARINE SCIENCE & RESEARCH

RESPONSIBLE WATER SOURCES. EFFICIENT SYSTEMS. HEALTHY ENVIRONMENT.

SUSTAINABLE SHADING SYSTEM

REDUCES SOLAR HEAT GAIN
Minimizes cooling loads through solar shading and reduces energy consumption.

OPTIMIZES DAYLIGHT
Allows natural light to penetrate while reducing glare and improving indoor comfort.

MARINE-INSPIRED DESIGN
Perforated patterns inspired by marine organisms enhance aerodynamic performance.

DURABLE & LOW MAINTENANCE
Corrosion-resistant materials ensure longevity in coastal environments.

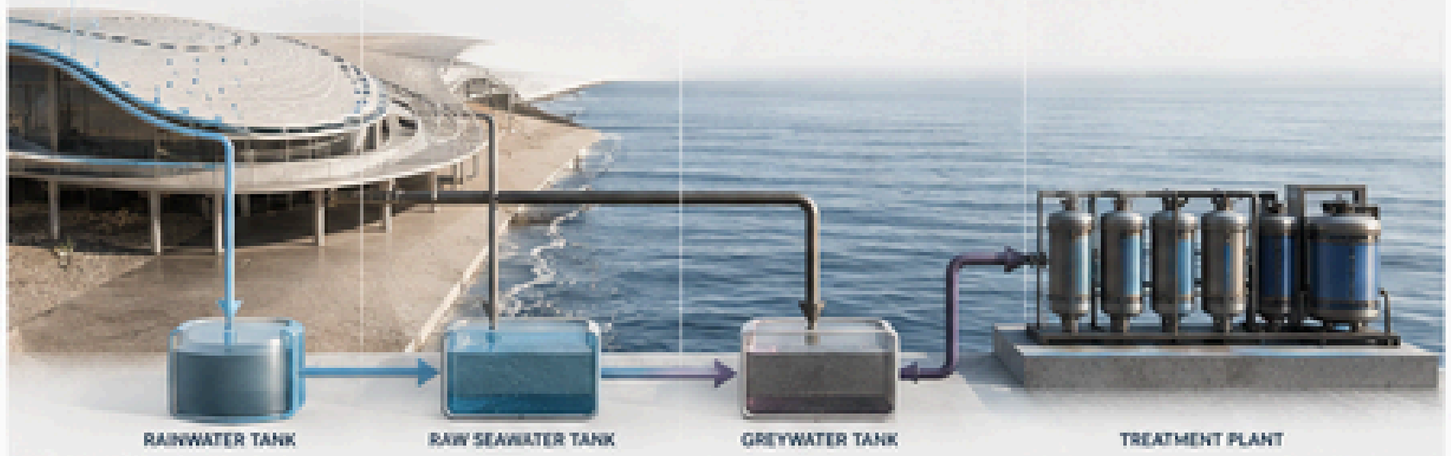
INTEGRATED WATER MANAGEMENT SYSTEM

RAINWATER HARVESTING
Rainwater collected from roofs and shaded surfaces.

SEAWATER INTAKE
Seawater is responsibly intaked from the sea.

GREYWATER REUSE
Greywater from sinks and showers is collected and treated.

WATER TREATMENT & FILTRATION
Advanced filtration ensures high-quality water for non-potable uses.



TREATED WATER APPLICATIONS

TOILET FLUSHING · LANDSCAPE IRRIGATION · COOLING SYSTEMS · AQUARIUMS & RESEARCH TANKS · CLEANING & MAINTENANCE

NATIVE LANDSCAPING & BIODIVERSITY

DROUGHT-TOLERANT PLANTS
Use of native species reduces irrigation needs and supports local ecosystems.

SOIL MOISTURE CONSERVATION
Smart irrigation and mulching techniques minimize evaporation and water waste.

ENHANCES MICROCLIMATE
Vegetation and water features cool the site naturally and improve air quality.

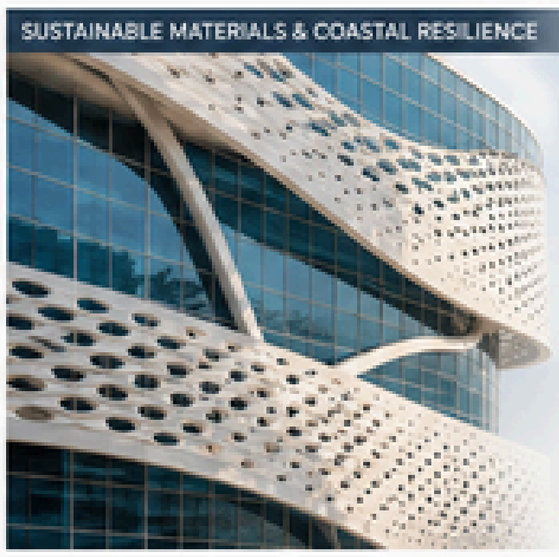
WATER FEATURES & MICROCLIMATE

EVAPORATIVE COOLING
Water features reduce ambient temperature through natural evaporation.

RELAXING ENVIRONMENT
Water elements create a calming atmosphere that enhances well-being and concentration.

STORMWATER MANAGEMENT
Surfaces are designed to manage and reuse stormwater efficiently.

SUSTAINABLE MATERIALS & COASTAL RESILIENCE



CORROSION-RESISTANT MATERIALS
Materials are selected to withstand harsh marine conditions.

LOW-IMPACT CONSTRUCTION
Local and recycled materials reduce the environmental footprint.

CLIMATE ADAPTIVE DESIGN
Building orientation and form enhance natural ventilation and reduce energy demand.

SUSTAINABLE WATER CYCLE

PRECIPITATION → COLLECTION → STORAGE → TREATMENT → REUSE → RETURN TO NATURE

SUSTAINABLE OUTCOMES

REDUCED WATER CONSUMPTION · LOWER ENERGY DEMAND · HEALTHIER ENVIRONMENT · SUPPORTS RESEARCH & EDUCATION · BUILT FOR THE FUTURE

لُجّة
LUJJA
WHERE KNOWLEDGE MEETS THE SEA

لُجّة

MAIN ENTRANCE

MAIN ENTRANCE

MARINA & BOAT BERTHS

MAIN ENTRANCE

AQUARIUM

WORKSHOPS

CLASSROOMS

MARINE RESEARCH LABORATORIES